DTS AND COMPUTER CONTROL

Annotations

Q1 This question was generally correctly done with a range of marks between 3 and 20 and an average of 12.67.
Some students did not remember the backward difference formula (although it is asked in the exam pretty consistently throughout the years). Note that memory is not necessarily required though: one student simply repeated the derivation from first principles and obtained the correct result.
Many students made computational mistakes. One rather silly computational mistake was made when Laplace-inverting the pole $\frac{1}{s-1}$, although the formula was provided in the formula sheet!
Few students recognized that the obtained poles are close because the sampling time is small.

Q2 This question was maybe too easy with the average mark of 15.33. Almost all students did points a, b and c correctly (some forgot to justify why the problem has a unique solution). Several students did not solve correctly points d and e. The most common error was to do a partial division, obtaining only 3 (incorrect) terms. Point e was very easy, requiring no computation, but yet there were too many incorrect answers.

Q3 This question was done well by most, with an average mark of 14.50. There were mostly computational mistakes in computing the reachability matrix, the transfer function and in determining the time T for which the system is not reachable/observable.

Q4 This question was perceived as the easiest. Errors were mostly done in computing the initial phase margin, which was -6. Many students wrote +6 or another number and consequently they picked a low m number (5 or 6) instead of 16.

Overall this exam was too easy and and the average was a few points over the target.

Answers

1. a) To take in account the effect of the zero-order hold, we consider open-loop transfer function

$$C(s)\widetilde{H}_0(s)P(s) = \frac{k_p s + 2}{s(s-1)(\frac{s}{20}+1)}.$$

The corresponding closed-loop characteristic polynomial is

$$s^3 + 19s^2 + (20K_p - 20)s + 40.$$

Applying the Routh criterion yields that the closed-loop system is asymptotically stable for $K_p > \dfrac{21}{19} \approx 1.1053$.

[4 marks]

b) The controllers are given by

$$C_B(z) = C(s)\Big|_{s=10\frac{z-1}{z}} = \frac{21z-20}{5z-5}$$

and

$$C_T(z) = C(s)\Big|_{s=20\frac{z-1}{z+1}} = \frac{41z-39}{10z-10}.$$

Both controllers have a pole at $z = 1$.

[4 marks]

c) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{1}{s(s-1)}\right) \\ &= \frac{e^{0.1}-1}{z-e^{0.1}}. \end{aligned}$$

[4 marks]

d) The poles of the closed-loop system are the roots of the characteristic polynomial

$$5z^2 + (16e^{0.1} - 26)z + 20 - 15e^{0.1},$$

namely

$$z_{B1} \approx 0.9171 \qquad z_{B2} \approx 0.7464.$$

[4 marks]

e) The poles of the two controllers are very close. The reason is because the sampling time is relatively small. If the sampling time were larger, then the two discretization algorithms would yield different controllers.

[4 marks]

Answers

2. a) By the properties of the z-transform we obtain

$$(1-2z^{-1}+z^{-2})Y(z) = (2z^{-1}+z^{-2})U(z)$$

which reordered yields

$$\frac{Y(z)}{U(z)} = \frac{1+2z}{(z-1)^2}.$$

[2 marks]

b) The problem has a unique solution if $m = n+p-1$, where m is the order of the denominator of the controller, n is the order of the denominator of the transfer function and p is the number of performance specifications. In this case, $p = 0$, $n = 2$ and $m = 1$, so the problem has a unique solution. The characteristic polynomial is

$$\begin{aligned} p(z) &= (z+v_1)(z-1)^2 + (s_0 z + s_1)(1+2z) \\ &= z^3 + (v_1 + 2s_0 - 2)z^2 + (1 - 2v_1 + s_0 + 2s_1)z + v_1 + s_1 \end{aligned}$$

and we want that $p(z)$ satisfies

$$p(z) = z^3.$$

This yields the system of equations

$$\begin{aligned} v_1 + 2s_0 - 2 &= 0, \\ 1 - 2v_1 + s_0 + 2s_1 &= 0, \\ v_1 + s_1 &= 0, \end{aligned}$$

which has solution $v_1 = \frac{4}{9}$, $s_0 = \frac{7}{9}$, $s_1 = -\frac{4}{9}$. Thus, the controller is

$$C(z) = \frac{7z-4}{9z+4}.$$

[6 marks]

c) The closed-loop transfer function is

$$\frac{Y(z)}{R(z)} = \frac{C(z)HP(z)}{1+C(z)HP(z)} = \frac{14z^2 - z - 4}{9z^3} = \frac{14}{9}z^{-1} - \frac{1}{9}z^{-2} - \frac{4}{9}z^{-3}.$$

[3 marks]

d) The reference signal $r(k) = 1$ has z-transform

$$R(z) = \frac{1}{1-z^{-1}} = 1 + z^{-1} + z^{-2} + z^{-3} + \ldots.$$

The z-transform of the output is

$$\begin{aligned} Y(z) = \frac{C(z)HP(z)}{1+C(z)HP(z)}R(z) &= \left(\frac{14}{9}z^{-1} - \frac{1}{9}z^{-2} - \frac{4}{9}z^{-3}\right)(1 + z^{-1} + z^{-2} + z^{-3} + \ldots) \\ &= \frac{14}{9}z^{-1} + \frac{13}{9}z^{-2} + z^{-3} + z^{-4} + \ldots \end{aligned}$$

Thus, the output sequence is

$$y(0) = 0, \qquad y(1) = \frac{14}{9}, \qquad y(2) = \frac{13}{9}, \qquad y(k) = 1, \text{ for } k \geq 3.$$

[6 marks]

Answers

e) Note that the open-loop transfer function has two poles in 1. Thus, the system is of type 2. Consequently:

i) For large k, $e(k) = 0$ because there is no steady-state error for constant or ramp references.

ii) For large k, $e(k) = \alpha$ with $\alpha \neq 0$. There is a finite steady-state error for parabolic references.

iii) $e(k) \to \infty$, as $k \to \infty$.

[3 marks]

Answers

3. a) As hinted, substituting $t = (k+1)T$ and $t_0 = kT$ in

$$x(t) = e^{F(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{F(t-\tau)}Gu(\tau)d\tau.$$

yields

$$x((k+1)T) = e^{FT}x(kT) + \int_{kT}^{(k+1)T} e^{F((k+1)T-\tau)}Gu(\tau)d\tau.$$

Notice that in the interval $[kT,(k+1)T)$, $u(t)$ is identically equal to $u(kT)$. Thus, it can be brought outside the integral. Again as hinted, using the change of variable $\lambda = (k+1)T - \tau$ yields

$$x((k+1)T) = e^{FT}x(kT) + \int_{T}^{0} e^{F\lambda}G(-d\lambda)u(kT),$$

from which the relations

$$A = e^{FT}, \qquad B = \int_0^T e^{F\lambda}Gd\lambda,$$

follow. The continuous-time equation for y is:

$$y(t) = Hx(t),$$

hence, for $t = kT$ we have

$$y(kT) = Hx(kT) = Cx(kT).$$

[4 marks]

b) To compute the matrices A and B we first compute $(sI-F)^{-1}$, namely

$$(sI-F)^{-1} = \begin{bmatrix} \dfrac{s}{s^2+1} & \dfrac{1}{s^2+1} \\ \dfrac{-1}{s^2+1} & \dfrac{s}{s^2+1} \end{bmatrix}.$$

It follows that

$$A = \begin{bmatrix} \cos(T) & \sin(T) \\ -\sin(T) & \cos(T) \end{bmatrix}, \qquad B = \int_0^T \begin{bmatrix} \sin(\lambda) \\ \cos(\lambda) \end{bmatrix} d\lambda = \begin{bmatrix} 1-\cos(T) \\ \sin(T) \end{bmatrix}.$$

[4 marks]

c) The reachability matrix of the discrete-time system is

$$P = [B \quad AB] = \begin{bmatrix} 1-\cos(T) & \cos(T)-\cos^2(T)+\sin^2(T) \\ \sin(T) & -\sin(T)+2\sin(T)\cos(T) \end{bmatrix},$$

the rank of which is less than 2 if $T = k\pi$. For instance, for $T = \pi$ we have

$$P = \begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix},$$

the rank of which is 1. Therefore the discrete-time system with $T = \pi$ is not reachable. For any $T \neq k\pi$, $k \in \mathbb{Z}_+$, the system is reachable.

[4 marks]

Answers

d) The observability matrix of the discrete-time system is

$$Q = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\sin(T) & \cos(T) \end{bmatrix},$$

the rank of which is less than 2 if $T = k\pi$. For instance, for $T = \pi$ we have

$$Q = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix},$$

the rank of which is 1. Therefore the discrete-time system with $T = \pi$ is not observable. For any $T \neq k\pi$, $k \in \mathbb{Z}_+$, the system is observable.

[4 marks]

e) To compute the transfer function we first determine the term

$$(zI - A)^{-1} = \begin{bmatrix} \dfrac{z - \cos(T)}{z^2 - 2\cos(T)z + 1} & \dfrac{\sin(T)}{z^2 - 2\cos(T)z + 1} \\ \dfrac{-\sin(T)}{z^2 - 2\cos(T)z + 1} & \dfrac{z - \cos(T)}{z^2 - 2\cos(T)z + 1} \end{bmatrix}.$$

The input-output transfer function is given by

$$\frac{Y(z)}{U(z)} = C(zI - A)^{-1}B = \frac{\sin(T)(z-1)}{z^2 - 2\cos(T)z + 1}.$$

For $T = k\pi$, $k \in \mathbb{Z}_+$, the transfer function is identically zero, because the system is unreachable and unobservable.

[4 marks]

Answers

4. a) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= \frac{z-1}{z} Z\left(\frac{P(s)}{s}\right) = \frac{z-1}{z} Z\left(\frac{1}{s^3}\right) \\ &= \frac{T^2(z+1)}{2(z-1)^2} = \frac{0.005(z+1)}{(z-1)^2}. \end{aligned}$$

[4 marks]

b) The transfer function in the w-plane is (recall that $T = 0.1$)

$$HP(w) = HP(z)|_{z=\left(\frac{1+0.05w}{1-0.05w}\right)} = \frac{1-0.05w}{w^2}.$$

[3 marks]

c) The acceleration constant in the w-plane is defined in the same way as in the s-plane. Thus,

$$K_a = \lim_{w\to 0} w^2\, C_1(w)\, HP(w) = K.$$

Selecting $K = 4$ yields $K_a = 4$. The open-loop transfer function is given by

$$C_1(w)HP(w) = \frac{4(1-0.05w)}{w^2}.$$

[3 marks]

d) The phase margin of the open-loop transfer function is approximately -6°. We need at least 56° more.

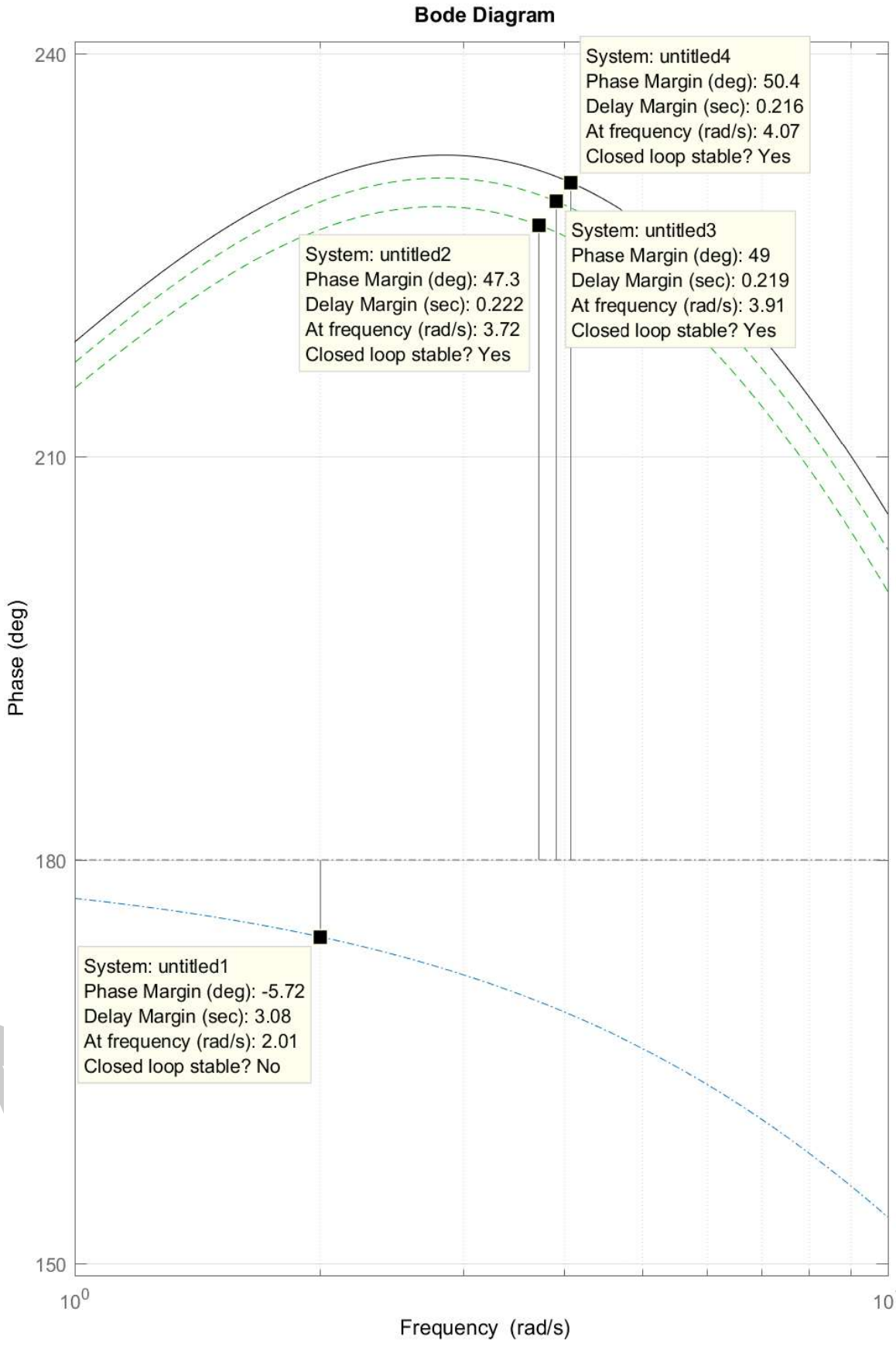
- We start selecting $m = 12$. The phase increase is approximately 58°. The magnitude decrease is approximately 11 dB. From the Bode plot, we read that -11 dB is approximately at $\omega = 3.8$ for which the phase margin is -11°. Since $58 - 11 = 47 < 50^\circ$, this compensator does not satisfy the phase margin requirement.
- We select $m = 14$. The phase increase is approximately 60°. The magnitude decrease is approximately 11.5 dB. From the Bode plot, we read that -11.5 dB is approximately at $\omega = 3.9$ for which the phase margin is -11°. Since $60 - 11 = 49 < 50^\circ$, this compensator does not satisfy the phase margin requirement.
- We select $m = 16$. The phase increase is approximately 62.5°. The magnitude decrease is approximately -12 dB. From the Bode plot, we read that -12 dB is approximately at $\omega = 4$ for which the phase margin is -11.5°. Since $62.5 - 11.5 = 51 > 50^\circ$, this is an acceptable solution.

Thus, for $m = 16$ and $\hat{\omega} = 4$, we have $\tau = \frac{\sqrt{16}}{4} = 1$ and the controller

$$C_2(w) = \frac{1+w}{1+\frac{1}{16}w}.$$

For illustrative purposes (*i.e.* this was not required) in the next page it is shown the Bode plot (phase) of the uncompensated system (dashed-dotted line), of the compensated system with $m = 12$ and $m = 14$ (dashed lines) and of the compensated system with $m = 16$ (solid line). [6 marks]

Answers


Bode Diagram
System: untitled4
Phase Margin (deg): 50.4
Delay Margin (sec): 0.216
At frequency (rad/s): 4.07
Closed loop stable? Yes
System: untitled3
Phase Margin (deg): 49
Delay Margin (sec): 0.219
At frequency (rad/s): 3.91
Closed loop stable? Yes
System: untitled2
Phase Margin (deg): 47.3
Delay Margin (sec): 0.222
At frequency (rad/s): 3.72
Closed loop stable? Yes
System: untitled1
Phase Margin (deg): -5.72
Delay Margin (sec): 3.08
At frequency (rad/s): 2.01
Closed loop stable? No
240
210
180
150
Phase (deg)
10^0
10^1
Frequency (rad/s)

e) The discrete-time controller is

$$C(z) = [C_1(w)C_2(w)]_{w=20\left(\frac{z-1}{z+1}\right)} = 37.333\frac{z-0.9048}{z-0.1111}.$$

Note that this is also equivalent to

$$C(z) = [C_1(w)]_{w=20\left(\frac{z-1}{z+1}\right)}\,[C_2(w)]_{w=20\left(\frac{z-1}{z+1}\right)}.$$

[4 marks]

Answers